

SAIDEEPTHI JAGARAPU

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EDUCATION

University of North Carolina at Charlotte, MS, Computer Science | CGPA : 4/4 Aug 2024 – Dec 2025

- **Relevant Coursework:** Algorithms & Data Structures, Software System Design & Impl, Network-Based App Development, Database Systems, Computer Networks, Distributed Systems, Cloud Computing, Big Data Analytics.

Indian Institute of Technology Guwahati, Bachelor of Technology, ECE July 2018 – May 2022

WORK EXPERIENCE

North Carolina Research Campus | Software Developer Nov 2024 – Present

- Implemented automated genome data ingestion pipelines using Java and Maven, integrating 220+ species and its huge datasets via REST APIs, leveraging asynchronous processing and caching for real-time data access.
- Developed genome visualization backend services in Java with Maven, refactoring legacy modules, implementing multithreading and optimized data structures, improving processing speed by 40% and data consistency by 30%.
- Deployed serverless and containerized pipelines on AWS ECS and Lambda, integrated with S3 and CloudWatch monitoring, enabling reliable, auto-scaled data workflows while reducing idle cloud resource costs by 20%.

Opportunity Knocks | Software Engineering Intern June 2025 – Aug 2025

- Collaborated with the founder to ship core features including real-time chat system, AI interview API integration, and Stripe payments using React, Typescript and Tailwind CSS, accelerating product rollout speed by 50%.
- Developed an AI Resume Evaluator using OpenAI API and prompt tuning, automating dynamic feedback while mentoring interns on architecture, code reviews, and UI/UX design, increasing evaluation throughput by 60%.
- Designed Supabase backend for 10K+ users with distributed architecture and role-based access, while integrating Hotjar analytics to monitor user behavior, identify bottlenecks, and improve system efficiency by 20%.

Walmart Global Tech India, Bangalore, India | Software Engineer II June 2022 – July 2024

- Architected 6 microservices using Java, Springboot REST APIs, implemented Redis caching, optimized complex queries, reducing backend latency by 25%, database load by 35% for applications serving 100k+ daily users.
- Migrated Java from JDK 8 to JDK 17 and upgraded Spring from 2.7 to 3.1, leveraging Java Streams, JPA repositories, enhancing security and system maintainability while reducing vulnerabilities by 40%.
- Refactored legacy monoliths into independent Java microservices, using multithreading, exception handling, circuit breakers, and centralized logging, improving throughput by 35% and system failure tolerance.
- Configured microservices with Kafka event streaming and Resilience4j Circuit Breaker and Retry patterns to enhance fault tolerance, prevent cascading failures, and ensure reliable asynchronous email/SMS notifications.
- Drove cross-functional team collaboration to develop React microfrontend applications and executed 25+ reusable Typescript+Angular UI components in the design library used by 20+ teams, increasing efficiency by 30%.
- Led performance optimization of large-scale distributed systems through thread-pool tuning, lazy loading, code splitting, and algorithm optimizations and caching, improving responsiveness and reducing page load time by 20%.
- Maintained CI/CD pipelines using Jenkins, Docker, and Kubernetes on GCP, integrating JUnit and Mockito-based automated testing to achieve 90% coverage and Grafana for monitoring, reducing deployment cycles by 20%.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript (ES6), TypeScript, HTML5, CSS3.

Frontend: React.js, Redux, Angular(RxJS, NgRx), Next.js (SSR, ISR), Bootstrap, jQuery, Webpack, Tailwind CSS, Jest.

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, Open-API standards, REST API, GraphQL, Node.js, FastAPI, Distributed Systems, Large-scale system design, Event Driven Microservice using Kafka.

Database & Tools: MySQL, MongoDB, PostgreSQL, Redis, Azure, Amazon Web Services(AWS), Terraform, Docker, Kubernetes, Jenkins, SDLC, Postman, Swagger, REST, Tomcat, Cursor, Claude, GenAI, Jira, Agile.

PROJECTS

AI Interview Coach (Capstone) | Stack: NextJS, Material UI, Supabase, GenAI Live Project

- Developed a full-stack AI-powered platform processing 1,000+ students and 5,000+ sessions, generating personalized quizzes and Q&A tailored to individual learning needs.
- Implemented dynamic content generation based on resumes, topics, and roles, achieving a 95% completion rate and significantly improving student engagement and learning outcomes.